

and it's time we do what we do best—tell you which one's right for you! We've tested 43 speakers in two categories—2.1 and the 4.1 / 5.1 (surround) group. The latter are better for movies, and the 2.1 range generally for music

German Physik Gaudi



The German Physik Gaudi is a \$2,50,000 speaker set. This speaker set comes with a choice of either titanium or carbon drivers. The weight of the cabinet is around 720 Kg. They are 2.1m in height and 1.3m in width. To keep the resonance at a minimum, there's use of a material called Hawaphon which is used widely in government and military buildings. If you wish, you can have a special carbon fibre cabinet.

and gaming. Let's find out then, shall we?

2.1 SPEAKER SYSTEMS

Multiple speakers don't result in louder and clearer sound. They especially don't make a lot of sense when you're going to spend your time listening to music more than anything else. 2.1 speakers have a few advantages of their own—for one, there's no trouble of setting up and arranging two or three additional speakers around you. Most surround speakers don't come with stands, so you need to make your own or hang them on a wall, which won't be at the right distance—so you end up with a completely ruined surround sound experience anyway. There are also cables to hide to make sure you don't have people tripping all over the place breaking their bones. However, if you want to use your system primarily for the home theatre movie experience, you'll need to spend on a 5.1 setup.

First Glimpse—Build Quality And Design

The focus on design and styling has become a big component of speakers, and all the speaker manufacturers are paying attention to it.

The Creative T3030 was an elegant-looking set with clear glossy plastic stands, which look very much like the other Inspire speaker sets. These comparatively flat and tall satellites are light, and the driver, a little high up, has enough weight to topple them over very easily. Another set of Creative speakers was on the other end of the spectrum—the M2600 satellites had a rather cheap plastic feel. They are meant to be thin and sleek too, but they have some provision made for the driver on the back.

TECH-COM in general looked to create very unique designs throughout the range of speakers they sent us. Although they had decent designs, the TECH-COM

SSD-803 was an ugly set, with spikes on four corners. It looks all right, but the front panels used on both the satellites and woofer look odd, are made of plastic, and aren't firmly attached. It was also kind of funny to see the exact same design used for the Intex IT-2050W and TECH-COM SSD-2000.

The Artis S111R and the S111 / FM were two models that look and sound quite the same, except that the latter has an FM radio and the other has a remote control. These, too, have decent build quality—the knobs had the right amount of resistance and sturdiness to them. The only issue was the pegs at the woofer, which make the woofer slide around a bit at higher volumes. The Zebronic ZEB-4000W's woofer was another sturdily-built set that looked simple yet attractive. The other Zebronic model—the ZEB-SW8000—has a very unattractive design, and there were dials on the front of the woofer with no indication of purpose.

One of the issues we looked at was the quality of cables provided by some of the speaker manufacturers. The Supercomp SES 2077, for example, comes with extremely thin cables which won't stand even a moderate tug. There were other problems, too: some of the TECH-COMs came with really short cables—so short with the SSD-815, that we were forced to keep the woofer on the table itself! Quite a few of the TECH-COM and Intex speakers used RCA connectors; we don't use sound cards, so we were forced to buy 3.5 mm-to-RCA converter cables. Some of the other TECH-COMs did come with the converter cables though.

There's something about large subwoofers that intimidates you, and the Altec Lansing MX5021 subwoofer clearly does that very well. It's huge, and easily the largest of all the 2.1 speakers. The other Altec Lansing woofers weren't all that great at times—especially the FX4021—with the down-firing setup. The panel on top of the woofer was really weak and is supported by just a metal mesh. The satellites are well designed and fairly sturdy. The Altec Lansing VS2421 has a very attractive remote with a nice red ring highlighting it.

It matched a similar design on the woofer and the satellites. Simple, good styling.

Quality Tests

The quality tests aren't just music tracks, but also recordings of all kinds of ambient background noises. Some of these were records with crackers exploding far in the distance and some right next to the recorder.

The Altec Lansing MX5021 came out far ahead of all the speakers in the 2.1 range. The ability to deliver clear sound at all volumes with very minor distortion at extreme volumes amazed us, and grabbed the attention of many passers-by. In one of the many tests, there was one where multiple bells were struck one after the other and it was amazing to hear the continuous hum from the different bells while new bells were added. Although the MX5021 isn't the warmest of speakers, the treble is still kept under check and doesn't go overboard.

The set leading the Logitech pack was the Z-2300. It was almost

Adam Professional Audio OSS



Adam's Professional Audio OSS has a total power output that adds up to around 3000W! It towers a mighty two metres in height and weighs well over 160 Kg. There are 60 drivers in this complete speaker set which are powered by the

12 amplifiers. The frequency range may be 18 Hz but it goes all the way up to 50 KHz—that's far beyond what humans can hear. That kind of performance should make animals on the other side of the globe go nuts. To keep it precise, there's a processor that makes changes to the delay to compensate for the location of the different drivers.

as powerful as the MX5021 but it lacked the prominent tight bass characteristic of the MX5021. If price is to be considered, it is more than a third costlier than the MX5021.

The Altec Lansing VS2421 was another set that wasn't as powerful as the MX5021, but clearly

How We Tested

Having received speakers from so many companies and for different setups, there were tests to gauge quality of the speakers and their performance under heavy load. The performance tests were basically a collection of tracks from different genres, as well as tracks designed to bring out fine details in speakers and highlight weaknesses.

All tests were performed using a Creative X-Fi Elite Pro sound card. Winamp was used to play the music, and Foobar for a few other tests. The bass and treble settings, wherever present, were kept at the default level.

Music is still where speakers are used the most, and that's why a big chunk of the tests involved listening to multiple tracks of different genres of music. Everything from Pop, Rock, Metal and Bollywood to Trance were used. Classical and vocalist-dependent tracks weren't missed out either. A *Sorta Fairytale* by Tory Amol and *Brid Og Ni Mhaille* by The Corrs were used. The Rock and Metal genre consisted of tracks from Opeth, Rush, Dream Theater, Break Benjamin, Between the Buried and Me and many more. The emphasis on these tracks was on the general sound and how each component of the track played out.

There were some quality tracks which were just ambient sounds like dogs bark-

ing and birds on the trees, water flowing down streams and so on. Each of these tracks had a list of events that was checked with every speaker—whether it could render these sounds properly. In some speakers, many of these events could not even be noticed. In the end, we ran the woofer and treble test with the volume cranked up to the maximum.

We also used Foobar to generate different frequencies, and then a frequency sweep of the whole spectrum to look for any kind of inaccuracy or inability to pump out certain frequency ranges. Many speakers show their glaring faults here.

For surround sound speakers, we used additional tests—HD video clips with high quality surround audio. These were clips from *Transformers*, which many consider to be one of the best CG movies ever. The fight scenes where metal-pounding action took place are where we focussed our attention. Bullets ricocheting off the robots and big bassy booms were also a good test to evaluate the 5.1 speakers. The same tests were used on the 2.1 speakers as well. Other than that, space fight scenes from *Star Wars: Revenge of the Sith* were used, and for ambient sounds, a few scenes from *Independence Day* were used. While the main dialogue between characters continues, there are background

sounds of cyclists riding and people walking, sounds of dice being rolled and the sounds of the waterfront next to which they're sitting—we wanted to see if the speakers could handle these well.

To evaluate surround sound, we use a benchmark called Rightmark 3D Sound. A looping track plays and keeps moving all around the listener—this is a great way to see how well surround sound systems perform. We looked for smooth transitions from one satellite to the next.

While all these tests were different from each other, half of the points were awarded to the general feel of sound and tone of the speaker and the other half were awarded on certain details present in each track. Tracks are played at different volumes to look for any signs of vibrations, fluttering or “tearing up”. In all, 40-50 different tracks were played on every single speaker set, trying to accommodate as many kinds of sounds and music genres as possible to get a good idea of what each of the speaker were capable of.

Other points were given to the build quality of the speakers, the stability of the stands and other features or effects that speakers might have. We also gave them scores on the type and quality of cables used, and the general usability of the features and effects.

How Do You Set Up A Surround Sound System?

It's one thing to set up a 5.1 speaker system, and it's a completely different thing to set it up right. It's understood that most people don't have the right furniture or the stands required to position their rear speakers suitably—so what do you do? Place them somewhere in the rear as long as they point to you. It's absolutely important that for a good surround sound experience, the distance, the direction and height of your satellites are just right. Compared to buying the speakers themselves, getting stands for the speakers is a minor cost, so it's well worth the money you spend for them.



The centre channel speaker should preferably be absolutely head-on to the people watching the movie. It should be placed at the centre of the screen, and in front of it. The front left and right satellites shouldn't be too close to the centre—if they are, you won't be able to make proper channel separation.

The same thing should be taken care of while setting up the rear speakers. The rear

speakers are where all the ambient sounds are projected from, so they will be pretty low in volume most of the time. The rear speakers can be much closer to viewers than the front channels.

The subwoofer—which generates the low frequency range in the sound and gives you the ground-shaking vibration—can be placed anywhere in the room, but for the right amount of volume of bass, you can choose

to keep it a little closer to you. Using the woofer next to the centre channel or next to the TV is also popular.

In the end, it all comes down to you. Play a few quiet movie scenes and some action packed

ones. Depending on the positioning of the speakers and the dimensions of the room, you'll need to adjust the volume for the rear speakers and also the distance.

All the satellite speakers should be placed on stands such that they are at the ear level of the audience—who could be on a sofa, for example. Also make sure you have all the speakers a foot or two away from the walls.